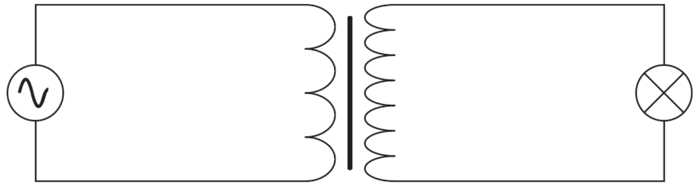
27. The primary coil of a step-up ideal transformer is connected to a source of alternating potential difference. The secondary coil is connected to a lamp.



Which line, **A** to **D**, in the table correctly describes the flux linkage and current through the secondary coil in relation to the primary coil?

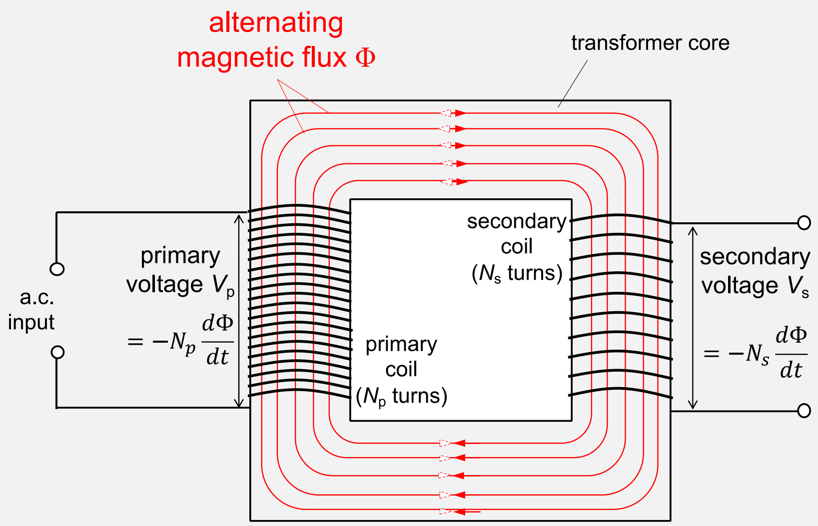
	$\dfrac{\text{secondary magnetic flux linkage}}{\text{primary magnetic flux linkage}}$	$\dfrac{\text{secondary current}}{\text{primary current}}$
A	> 1	< 1
B	< 1	< 1
C	> 1	> 1
D	< 1	> 1

Answer: A

$$\frac{N_S}{N_P} = \frac{V_S}{V_P} = \frac{I_P}{I_S}$$

For a step-up transformer, $N_S > N_P$. So, $V_S > V_P$ and $I_P > I_S$

Consider the ideal transformer,



- alternating current in primary coil gives rise to an <u>alternating magnetic flux</u> Φ in the core <u>in phase</u> with the current in the primary coil
- since the flux is changing, by law of EMI, an e.m.f. will be induced in the primary coil is $V_P = -N_P \dfrac{d\Phi}{dt}$ and the secondary coil is $V_S = -N_S \dfrac{d\Phi}{dt}$
- so, $V_P = -\dfrac{d}{dt}(N_P\Phi)$ and $V_S = -\dfrac{d}{dt}(N_S\Phi)$ where $N\Phi$ is flux linkage

- Since $V_S > V_P$, flux linkage of secondary coil > primary coil (see below)